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Goal

- Frontier language models demand enormous compute and can suffer unstable sparse routing or expensive communication.
- DeepSeek-V3 seeks closed-model-class capability with an open MoE trained at substantially lower cost.

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Training systems

MTP, FP8 AND DUALPIPE

- Multi-token prediction trains each position to forecast additional future tokens, enriching supervision and enabling speculative decoding.
- FP8 mixed-precision training and DualPipe pipeline scheduling reduce memory, communication bubbles and cost.

2

Scale

671B TOTAL, 37B ACTIVE

- The model has 671B total parameters and activates 37B per token using DeepSeekMoE and Multi-head Latent Attention.
- Pretraining uses 14.8T diverse high-quality tokens, followed by supervised fine-tuning and reinforcement learning.

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Evidence

2.788M H800 GPU HOURS

- Full training consumes 2.788 million H800 GPU hours; the report records no irrecoverable loss spikes and no rollbacks.
- Evaluations place DeepSeek-V3 above other open models and comparable with leading closed models across broad tasks.

3

Routing innovation

NO AUXILIARY BALANCING LOSS

- A bias-based expert load-balancing strategy adjusts routing dynamically without adding a performance-harming auxiliary loss.
- A small sequence-wise balance term remains for extreme cases while expert specialization is preserved.

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Meaning & limits

- V3 demonstrates that sparse architecture, routing objectives, prediction targets and hardware scheduling can jointly unlock scale.
- Training still needs major infrastructure; contamination, data governance, MoE serving complexity and safety require independent study.

DEEPSEEK-V3

DeepSeek-V3 Technical Report

DeepSeek-AI | arXiv:2412.19437v2

Scale the efficient MLA-MoE stack to 671B parameters while improving routing, token prediction and low-precision systems for stable, economical frontier training.

671B MoE

aux-loss-free routing

FP8 + MTP